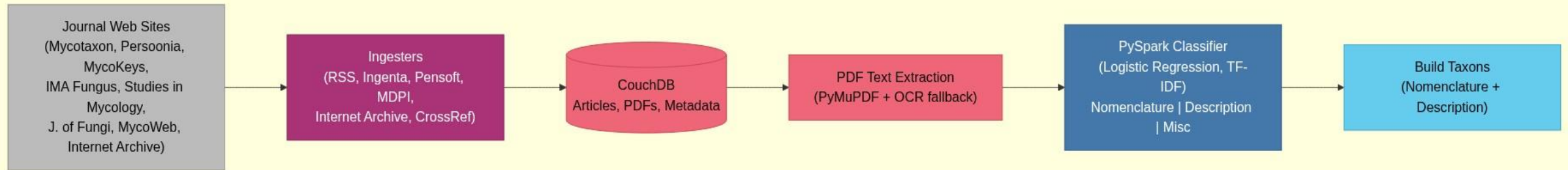


Building the Synoptic Key of Life website

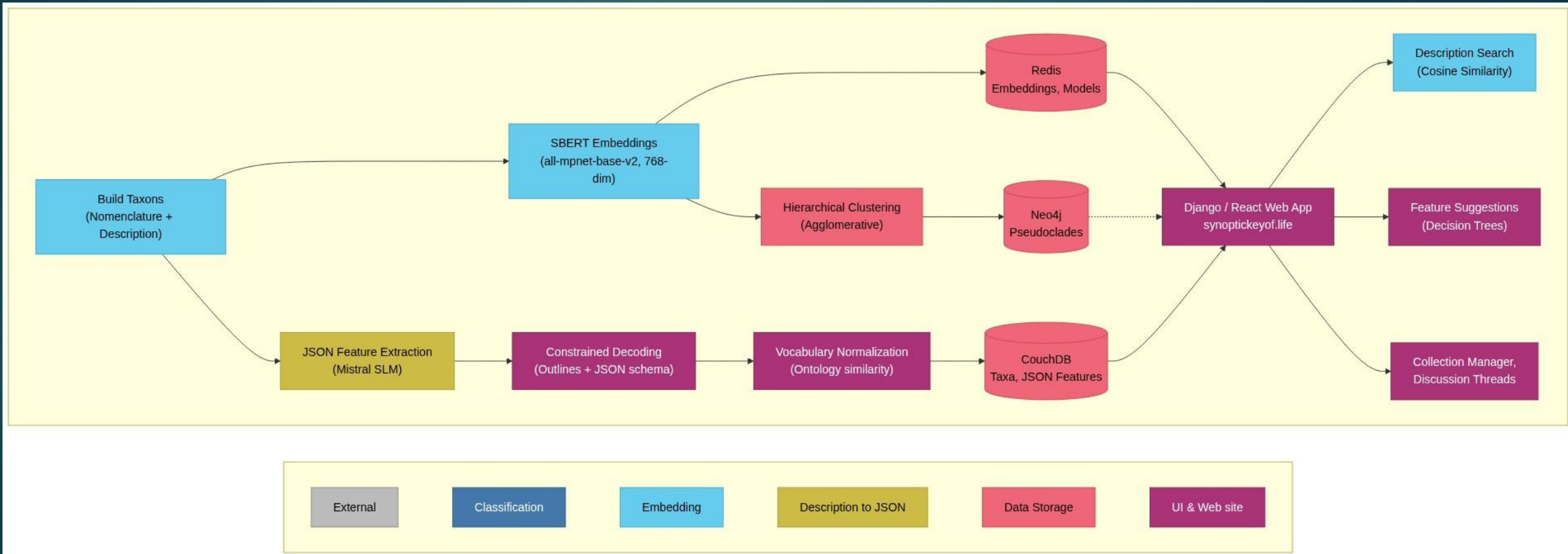
- ▶ La Monte Henry Piggy Yarroll
- ▶ MASMC 2026
- ▶ 2026-04-25
- ▶ North Carolina Agricultural and Technical State
(NCA&T) University



SKOL Architecture by Implementation Project (1 of 2)



SKOL Architecture by Implementation Project (2 of 2)

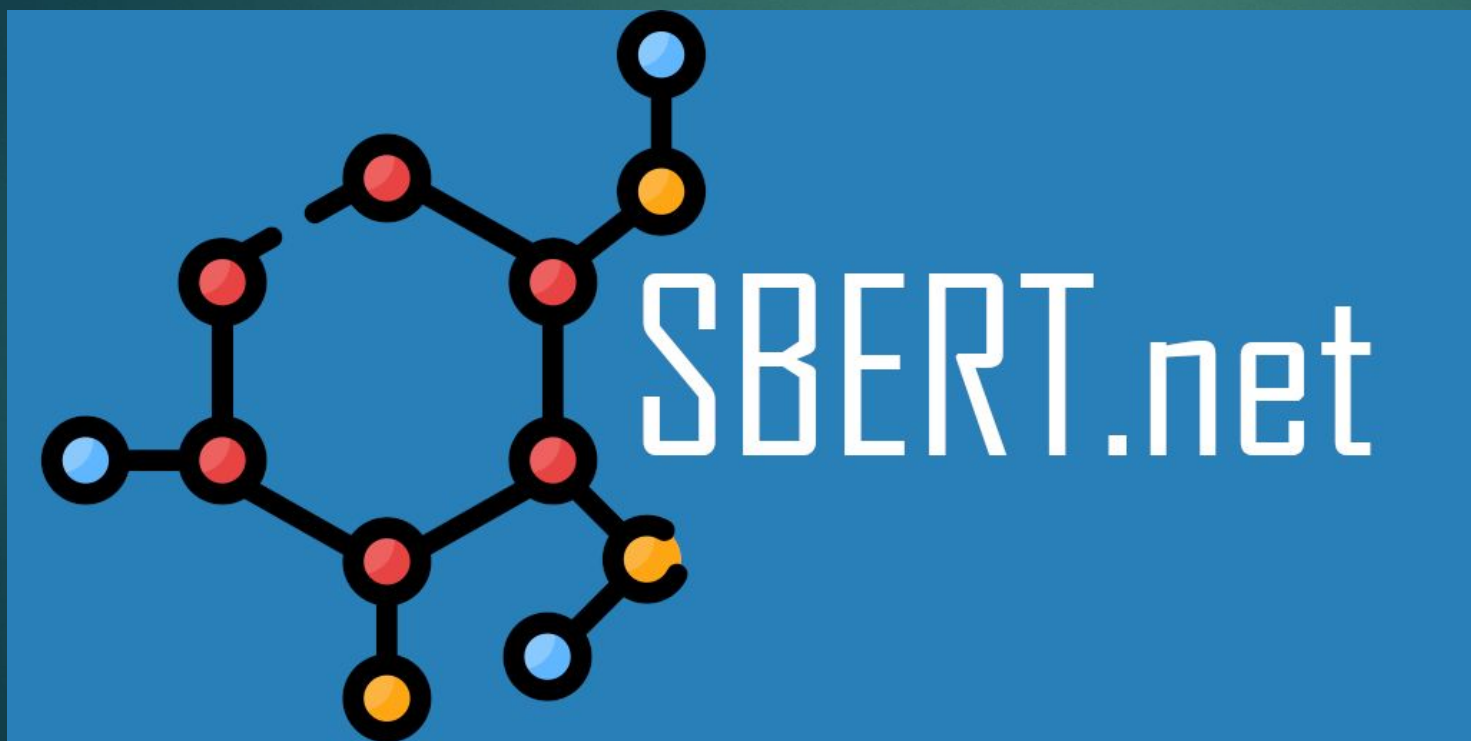


YEDDA Annotation (Classification)

[@2. *Vibrissea filisporia* (Bonord.) Korf & A. Sánchez f.
boudieri A. Sánchez & Korf in Sánchez, J. Agric.
Univ. Puerto Rico 51: 83, 1967.
FIGS. 3, 4 & 5.#Nomenclature*]

[@APOTHECIA sessile, discoid, 0.75-1 mm in diameter.
RECEPTACLE brown. DISC convex, buff when rehydrated.
HYMENIUM making up to 3/4 of the apothecium, asci taking up
most of the height. ECTAL EXCIPULUM brown, 65.3-76.4 µm
wide, of textura globulosa, with cells turning
outwards#Description*]

Dr Draft's state-of-the-art (SOTA) Literature Search



Embedding

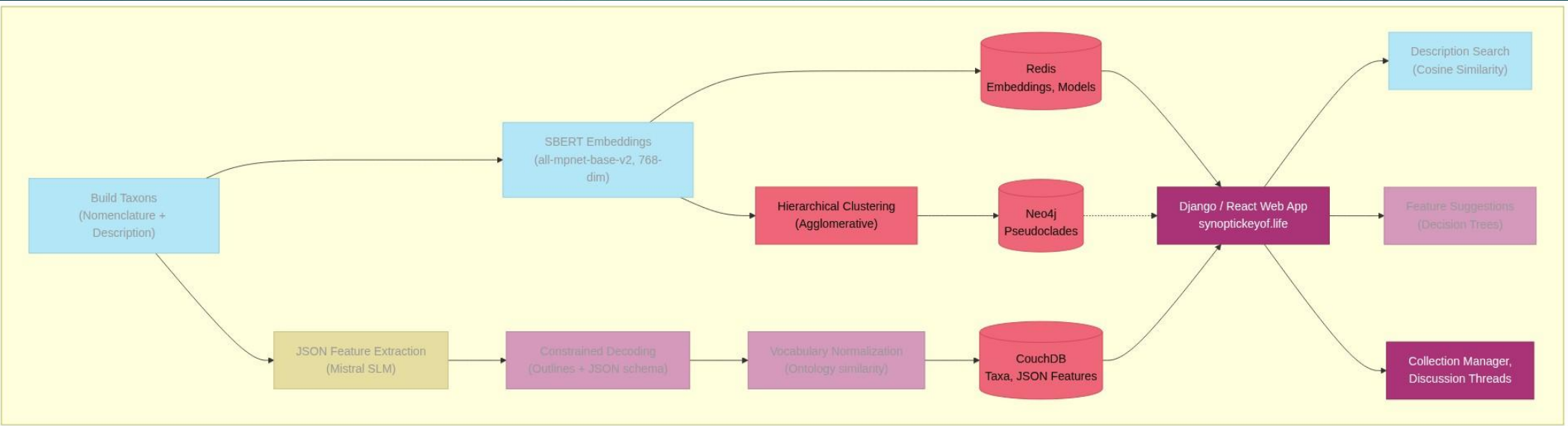
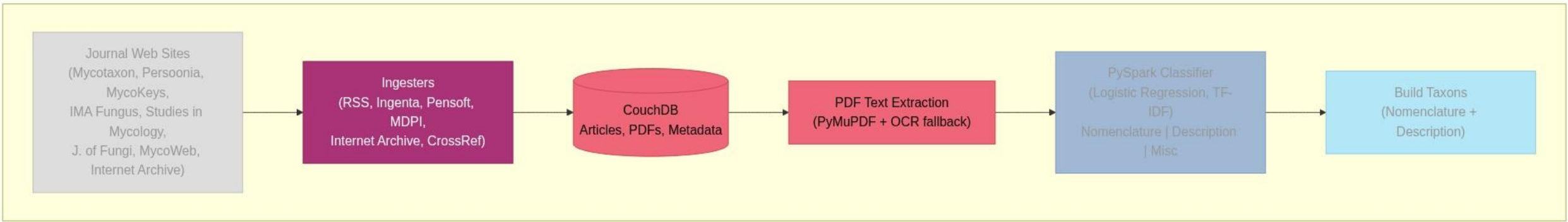
Description to JSON

Can you extract any attributes and their measurements from the following description of an organism? Present the result as a json data structure.

Mycelium on the substrate is medium orange-brown, septate, 3-4 μm diam., often distinctly and coarsely rough; hyphae often run parallel to the long axis of cells in substrate tissue but usually branch on the exterior and appear as a non-patterned network. Sporulation units are produced as enlarging tips of short branches, at first concolorous with surface mycelium but soon dark orange-brown and finally black-brown and opaque. In type material an occasional enlarging branch can be seen attached to the hypha from which it developed; in most instances, however, the enlarged branch elements have broken loose from the mycelium or are so densely crowded on the substrate and so nearly opaque that points of attachment cannot be traced

```
{
  "mycelium": {
    "location": "on substrate",
    "color": "medium orange-brown"
  },
  "hyphae": {
    "septate": true,
    "diameter_um": {
      "min": 3,
      "max": 4
    },
    "surface": "distinctly and coarsely rough",
    "orientation": "parallel to long axis of substrate tissue cells",
    "branching": {
      "location": "on exterior",
      "pattern": "non-patterned network"
    }
  },
  "sporulation_units": {
    "origin": "enlarging tips of short branches",
    "color_progression": [
      "concolorous with surface mycelium",
      "dark orange-brown",
      "black-brown and opaque"
    ],
    "attachment": {
      "occasional": "attached to hypha",
      "typical": "broken loose or obscured by crowding and opacity"
    }
  }
}
```

Description to JSON



Pseudoclades

```
neo4j$ match(c1:Pseudoclade)-[p:PARENT_OF]→(c2) return c1, p, c2
```

Taxon

<elementId>	4:682b3416-e2f6-408a-bb78-4f8ab353470e:1309
<id>	1309
description	Starch-like substances are not produced. Growth on 50 % (w/w)
empirical_page_number	24
line_number	14203
name	Vankyiozyma motuoensis Q.-M. Wang, sp. nov.
node_id	255
page_number	1
paragraph_number	19127
source_db_name	skol_dev
source_doc_id	b41960c88bd54801b0a819e98682d542
source_human_url	skol_dev/b41960c88bd54801b0a819e98682d542/article.txt.ann

```
neo4j$ match(c1:Pseudoclade)-[p:PARENT_OF]→(c2) return c1, p, c2
```

Taxon

<elementId>	4:682b3416-e2f6-408a-bb78-4f8ab353470e:1310
<id>	1310
description	Starch-like substances are not produced. Growth on 50 % (w/w)
empirical_page_number	24
line_number	14065
name	Exobasidium lijiangense Q.-M. Wang, sp. nov.
node_id	1207
page_number	1
paragraph_number	19115
source_db_name	skol_dev
source_doc_id	b41960c88bd54801b0a819e98682d542
source_human_url	skol_dev/b41960c88bd54801b0a819e98682d542/article.txt.ann

Search Page (showing JSON Feature Tool)

Welcome, niemand

My Collections

Help

Feedback

Contact

Account

Logout

SKOL Semantic Search α

Search taxonomic descriptions using semantic similarity

Try this:

campanulate pileus, stipe stains blue when cut

Active Collection: inat stereum x v

+ New

ID: 601199613 [View Details](#)

Nomenclature: Unknown

Description Text ([linked to collection](#)) [Flag](#)

orange margin; hyphaesystem ketoatocalhyphae thickened walls; spores (7-) 10.5 - 11.3 (-12) $\times$ (6.5-) 9.5 - 10 (-11) μm , Q: (1-)1.1-1.1(1.1); paraphyses (100-) 116 - 137 (-147) μm ;

Vocabulary

Text Features

JSON Features

Metrics

100 taxa, 739 features, depth 10

☐ conidia cylindrical shape 0 cylindrical

☐ growthform 11 description 0 epihymenium colourless, faintly yellow

☐ meissner hydnoidea arrangement 0 opposite or alternate

☐ conidioma structure 1 1-4 mm diam.

Search Results

Search

Leucocoprinus

List Names

Discussion (0)

External Identifiers

GenBank

123456

Delete

Select type...

Identifier value (e.g., 336010515)

Add

Found 22 results

From collection search: "pileus form umbo low central umbo; pileus microstr..."

Loading 22 results from collection...

From: pileus form umbo low central umbo; pileus microstr...

Leucocoprinus tenellus Pegler [nom. illegit., non (Boud.) Locq.]

Context

97.3%

Label Collection

Feed: User Collection

pileus form umbo low central umbo; pileus microstructure pileipellisform cutis;asci spores per ascus 4;spores (11-) 13 - 13 (-14) × (9.5-) 10 - 10 (-11) μm, Q: (1.1-)1.3-1.3(1.4);

Show Full JSON

Leucocoprinus tenellus Pegler [nom. illegit., non (Boud.) Locq.] from the

Context

82.6%

Label Collection

Feed: CouchDB Taxa PDF Page: 7

4b. Pileus white to yellowish-brown, pale brown or greyish; stipe whitish, without reddish, greenish or bluish tinges; spores 5–8 μm in broad. 5

5a. Pileus 15–25 mm in diam., hemispheric to convex, pale brown with

stipe greyish white to yellowish, smooth or covered by

View PDF

View Source

Show Full JSON

Vocabulary Feature Tool

Vocabulary

Text Features

JSON Features

Metrics

Add Vocabulary Term

Clear

Pileus

×

▼

↳

Color

×

▼

↳

Pale brown

×

▼

Preview:

pileus color pale brown;

Add to Description

Text and JSON Feature tools

Vocabulary **Text Features** JSON Features Metrics

22 taxa, 5000 features, depth 10

☐	nom illegit		100.0%
☐	as am		100.0%
☐	1977		100.0%
☒	almost translucent		100.0%
☐	covering consists		100.0%
☐	fuscis in		100.0%
☐	changed colour		100.0%
☐	prague 48		100.0%
☐	basidia 14		100.0%

Preview: almost translucent;

Add to Description

Vocabulary Text Features **JSON Features** Metrics

22 taxa, 356 features, depth 6

☒	odor absent to sometimes fungal to distinctly fungal 2 distinctly
☐	physical characteristics sporal morphology spores other proper
☐	perithecium location country 0 in djur regione
☐	species basidia abnormal sterigmata length 0 up to \$16 \$µm lo
☐	speciesname morphology annulus color 0 white
☒	mushroom lamellae 1 thin

Preview: odor absent to sometimes fungal to d.

Add to Description

Metrics Description Tool

Vocabulary Text Features JSON Features Metrics

Feature: spores (30) ▼

spores (7-) 11 - 12 (-13) × (6.5-) 10 - 10
(-11) μm, Q: (1-)1.1-1.2(1.3)

n = 30

Dimensions: ☒ 2D ☐ 1D

L × W μm +

☒ Report Q values

Samples:

▼

Delete "spores" set

Add to Description

Account and Settings

[Feedback](#) [Contact](#) [Account](#) [Logout](#) 

Change Password

Connected Accounts

Link social accounts to sign in with them

 Google 112079383051265085039 [Disconnect](#)

 GitHub 7216308 [Disconnect](#)

 ORCID 0009-0006-3073-6352 [Disconnect](#)

Connect New Account

 Connect GitHub

 Connect Google

 Connect ORCID

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Search Settings

Embedding Model

skol:embedding:v1.1 

Total Results (k)

100

Results Per Page

10

Nomenclature Results

20

Feature Settings

Taxa for Features

6

Feature Count

30

Tree Depth

10

Min Doc Frequency

1

Max Doc Frequency

1

Email Preferences

☒ Receive daily admin summary

UI & Web site

Future

- Currently in alpha-test
- Data quality project, especially for older material
- Glossary tools
- Better multi-language support
- More journals and books
- Expand to other groups of organisms
- See <https://trello.com/b/jUkRE6zv/skol> for a complete backlog.



Acknowledgements

Development team

- Dr. Gregory Block (Faculty advisor)
- Christopher Murphy
- Jennifer Balasi
- Shintaro Osuga
- David Caspers
- Padmaja Kurumaddali
- Patrick Le

Alpha testers

- Garrett Taylor
- Richard Jacob
- Cara Coulter



These slides:

<https://tinyurl.com/ykwt8dkv>

The accompanying paper:

<https://tinyurl.com/54paunua>

The SKOL web site:

<https://synoptickeyof.life/skol>

The SKOL source code:

<https://github.com/piggyatbaqaqi/skol>

Dr Draft's state-of-the-art (SOTA) Literature Search

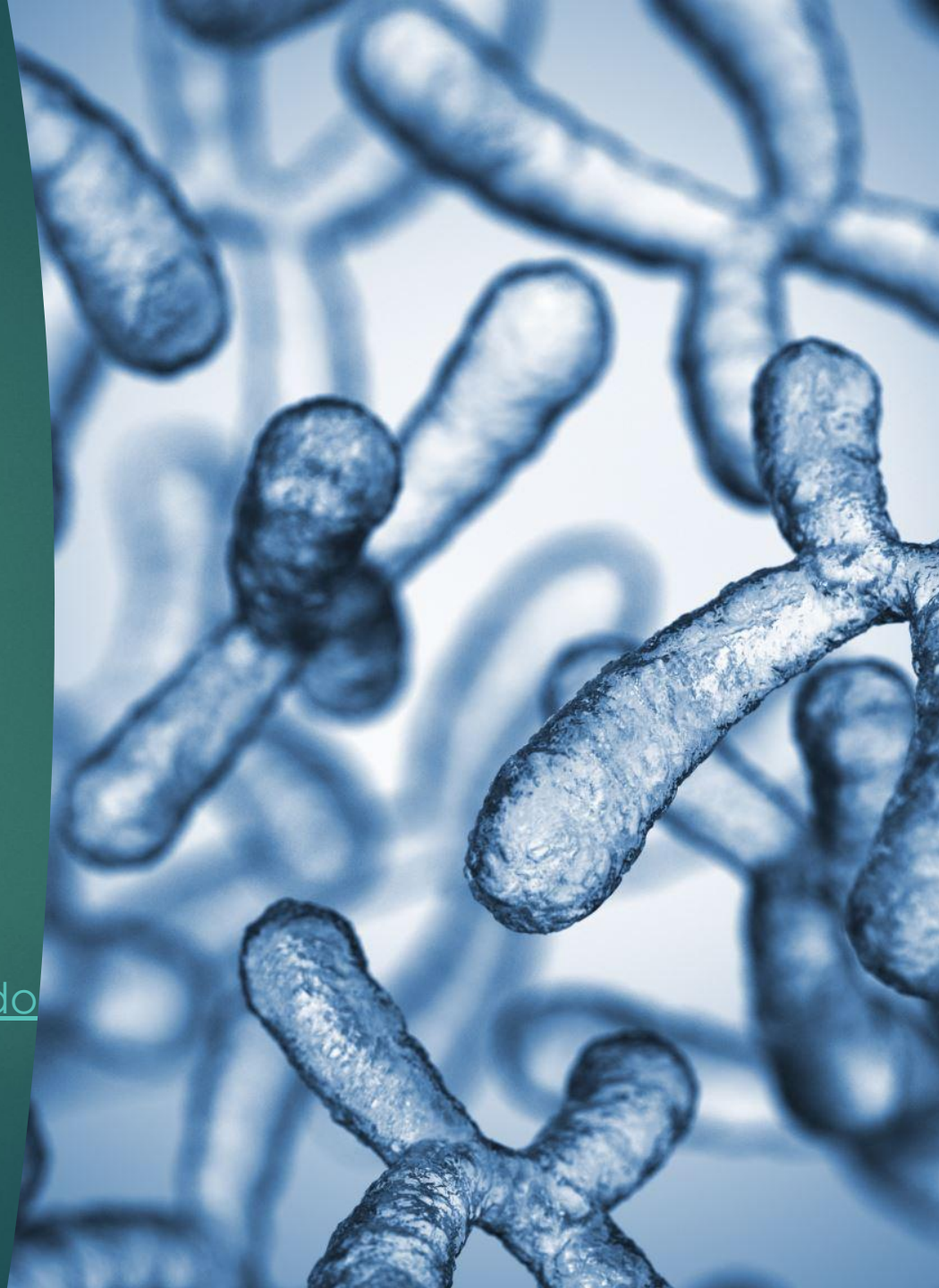
<https://github.com/autonlab/dr-drafts-sota-literature-search>

SKOL fork of Dr Draft's:

<https://github.com/piggyatbaqaqi/dr-drafts-mycosearch>

A complete summary of technologies integrated:

<https://github.com/piggyatbaqaqi/skol/blob/main/do>



Western Pennsylvania Mushroom Club SCHOLARSHIP/GRANT PROGRAM



**EDUCATIONAL
SCHOLARSHIP:**
Support for academic
studies in mycology or
related fields.



PROJECT GRANT:
Funds for a specific
mycological project,
research, or conservation.

HOW TO APPLY



Scan to Apply Online
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education/wpme-scholarship-
grant-application/](https://wpamushroomclub.org/education/wpme-scholarship-grant-application/)



Apply via Email:
schoiarship@wpamushroomclub.org

Fungi, fun and friends!

Appendix: Sources

Publication	Publisher/Source	Website	Total Documents	Taxonomy Documents	Taxa Documents
Cryptogamie, Mycologie	crossref	https://sciencepress.mnhn.f...	987	0 (0.0%)	8
Fungal Systematics and Evolution	ingenta	https://api.ingentaconnect.c...	139	0 (0.0%)	298
Journal of Fungi	mdpi	https://www.mdpi.com/journ...	5039	0 (0.0%)	387
Mycologia (Taylor & Francis)	crossref	https://www.tandfonline.co...	5078	0 (0.0%)	2
Mycology: An International Journal on Fungal Biology (Taylor & Francis)	crossref	https://www.tandfonline.co...	508	0 (0.0%)	13
Mycology: An International Journal on Fungal Biology (Taylor & Francis)	Unknown	—	15	0 (0.0%)	0
Mycoscience: An open access journal of the Mycological Society of Japan	crossref	https://mycoscience.org	2027	0 (0.0%)	39
Mycotaxon	ingenta	https://api.ingentaconnect.c...	2199	1881 (85.5%)	2002
Open Access Journal of Mycology & Mycological Sciences	medwin-publishers	https://www.medwinpublish...	112	0 (0.0%)	3
Persoonia	ingenta	https://api.ingentaconnect.c...	161	0 (0.0%)	1035
Studies in Mycology	ingenta	https://api.ingentaconnect.c...	53	45 (84.9%)	731
Sydowia (Internet Archive)	internet-archive	https://archive.org/details/p...	12	0 (0.0%)	27
Unknown	Unknown	—	7541	0 (0.0%)	15575